CO3 ASSESSMENT TOOL 1

Experiments-Based Assessment

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Source code(C Program - Merge Sort Performance Analysis)

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>


void merge(int arr[], int left, int mid, int right)

{

    int i, j, k;

    int n1 = mid - left + 1;

    int n2 = right - mid;


    int *L = (int *)malloc(n1 * sizeof(int));

    int *R = (int *)malloc(n2 * sizeof(int));


    for (i = 0; i < n1; i++)

        L[i] = arr[left + i];


    for (j = 0; j < n2; j++)

        R[j] = arr[mid + 1 + j];


    i = 0;

    j = 0;
```

```
    k = left;

    while (i < n1 && j < n2)
    {
        if (L[i] <= R[j])
            arr[k++] = L[i++];
        else
            arr[k++] = R[j++];
    }

    while (i < n1)
        arr[k++] = L[i++];

    while (j < n2)
        arr[k++] = R[j++];

    free(L);
    free(R);
}

void mergeSort(int arr[], int left, int right)
{
    if (left < right)
    {
        int mid = left + (right - left) / 2;

        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);

        merge(arr, left, mid, right);
```

```c
    }
}

int main()
{
    int n = 100000;

    int *arr = (int *)malloc(n * sizeof(int));

    srand(time(NULL));

    for (int i = 0; i < n; i++)
        arr[i] = rand();

    clock_t start = clock();

    mergeSort(arr, 0, n - 1);

    clock_t end = clock();

    double time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("Array Size : %d\n", n);
    printf("Execution Time : %lf seconds\n", time_taken);

    printf("\nFirst 10 Sorted Elements:\n");

    for (int i = 0; i < 10; i++)
        printf("%d ", arr[i]);
```

```
    free(arr);

    return 0;
}
```